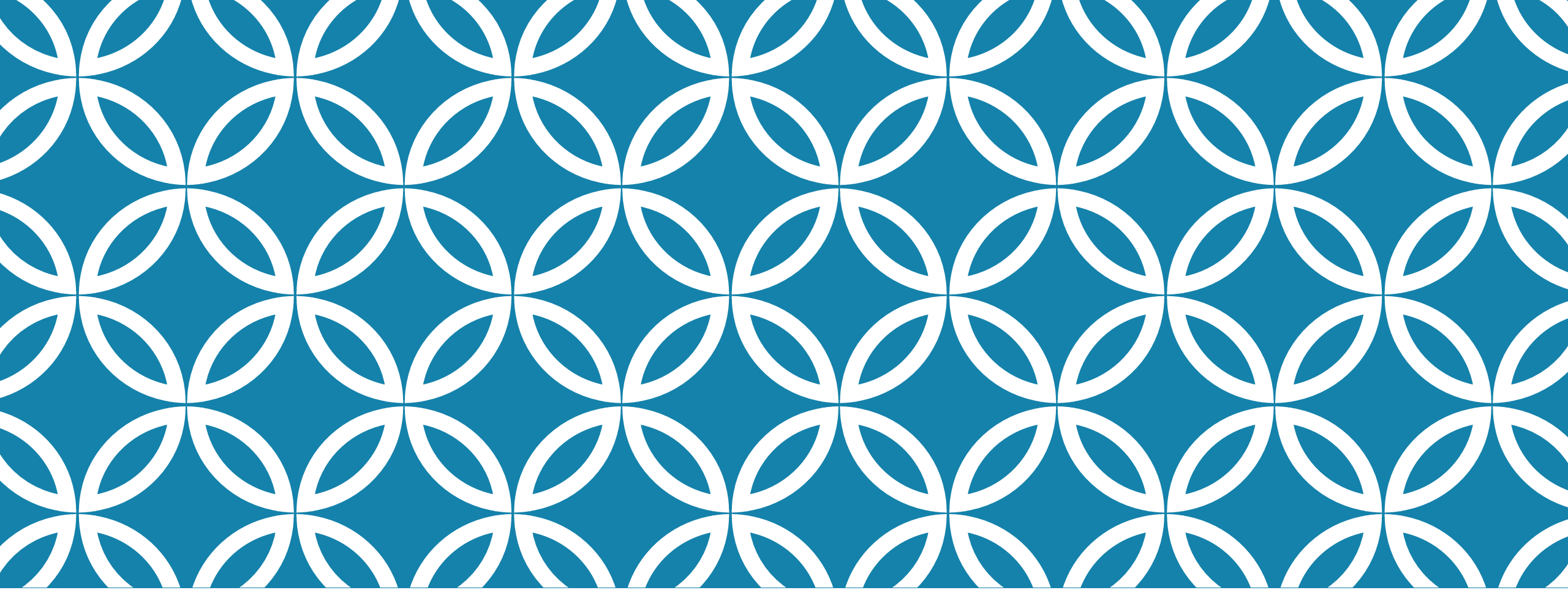


ARTIFICIAL INTELLIGENCE

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ABOUT COURSE-

Course Objectives –

1. Understand the fundamental concepts, terminology, and history of artificial intelligence. Understand the concept of state space representation and how it applies to problem-solving in AI.
2. Understand different methods for representing knowledge, including logic, semantic networks, frames, and ontologies**.**

Course Outcomes –

CO1. Understanding of the concepts and principles of artificial intelligence, including problem-solving, and intelligent agents.

CO2. Understand the concept of state space representation and how it applies to problem-solving in AI.

CO3. Analyze and apply different search techniques to solve problems, including uninformed search.

CO4. Understand the various methods of representing knowledge in AI systems, including propositional and predicate logic, semantic networks, and frames.

CO5. Understand the process of acquiring knowledge from domain experts and structuring it effectively in a knowledge base.

UNIT – I

Topics to be covered –

- Introduction, Overview of Artificial intelligence: Problems of AI, AI technique, Tic - Tac - Toe problem.
- Intelligent Agents, Agents & environment, nature of environment, structure of agents, goal based agents, utility based agents, learning agents.

WHAT IS AI ?

- It is a branch of Computer Science that pursues creating the computers or machines as intelligent as human beings.
- It is the science and engineering of making intelligent machines, especially intelligent computer programs.

Definition: Artificial Intelligence is the study of how to make computers do things, which, at the moment, people do better.

According to the father of Artificial Intelligence, **John McCarthy**, it is ***"The science and engineering of making intelligent machines, especially intelligent computer programs"***

- AI is accomplished by studying how human brain thinks and how humans learn, decide, and work while trying to solve a problem, and then using the outcomes of this study as a basis of developing intelligent software and systems.
- From a business perspective AI is a set of very powerful tools, and methodologies for using those tools to solve business problems.
- From a programming perspective, AI includes the study of symbolic programming, problem solving, and search.

PROBLEMS OF AI:

Before jumping on study of specific AI problems and solution techniques , it is important at least to discuss, if not to answer, the following four questions:

1. What are the underlying assumptions about intelligence?
2. What kinds of techniques will be useful for solving AI problems?
3. At what level human intelligence can be modelled?
4. When will it be realized when an intelligent program has been built?

AI TECHNIQUES

Artificial Intelligence research during the last three decades has concluded that Intelligence requires knowledge. To compensate overwhelming quality, **knowledge** possesses less desirable properties.

A. It is huge.

B. It is difficult to characterize correctly.

C. It is constantly varying.

D. It differs from data by being organized in a way that corresponds to its application.

E. It is complicated.

AN AI TECHNIQUE IS A METHOD THAT EXPLOITS KNOWLEDGE THAT IS REPRESENTED SO THAT:

- The knowledge captures generalizations that share properties, are grouped together, rather than being allowed separate representation.
- It can be understood by people who must provide it—even though for many programs bulk of the data comes automatically from readings.
- In many AI domains, how the people understand the same people must supply the knowledge to a program.
- It can be easily modified to correct errors and reflect changes in real conditions.
- It can be widely used even if it is incomplete or inaccurate.
- It can be used to help overcome its own sheer bulk by helping to narrow the range of possibilities that must be usually considered.

EXAMPLE – QUESTION ANSWERING PROGRAM

QUESTION --- Russia massed troops on the Czech border.

Following question-answering dialogues did occur with the **POLITICS** program [Carbonell, 1980]):

Dialogue 1

Q: Why did Russia do this?

A: Because Russia thought that it could take political control of Czechoslovakia by sending troops.

Q: What should the United States do?

A: The United States should intervene militarily.

Dialogue 2

Q: Why did Russia do this?

A: Because Russia wanted to increase its political influence over Czechoslovakia.

Q: What should the United States do?

A: The United States should denounce the Russian action in the United Nations.

APPROACH

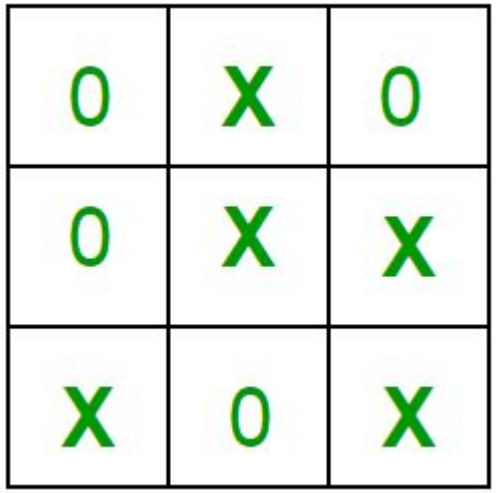
In the **POLITICS** program, answers were constructed by considering both the input text and a separate model of the beliefs and actions of **various political entities**, including Russia. When the model is changed, as it was between these two dialogues, the system's answers also change. In this example, the first dialogue was produced when POLITICS was given a model that was intended to correspond to the beliefs of a typical **American conservative** (circa 1977). The second dialogue occurred when POLITICS was given a model that was intended to correspond to the beliefs of a typical **American liberal** (of the same vintage).

The general point here is that defining what it means to produce a correct answer to a question may be very hard. **Usually, question-answering programs define what it means to be an answer by the procedure that is used to compute the answer**. Then their authors appeal to other people to agree that the answers found by the program "make sense" and so to confirm the model of question answering defined in the program. This is not completely satisfactory, but no better way of defining the problem has yet been found.

TIC-TAC-TOE GAME

Rules of the Game

- The game is to be played between two people (in this program between HUMAN and COMPUTER).
- One of the player chooses ‘O’ and the other ‘X’ to mark their respective cells.
- The game starts with one of the players and the game ends when one of the players has one whole row/ column/ diagonal filled with his/her respective character (‘O’ or ‘X’).
- If no one wins, then the game is said to be draw.



MINIMAX ALGORITHM?

Minimax is usually a decision-making algorithm for two-player games (like chess, tic-tac-toe, or checkers). It allows computers to plan several moves while assuming your opponent will always make their best possible move.

The Minimax algorithm gets its name from how it works:

- “Mini” represents the opponent trying to minimize your score
- “Max” represents you trying to maximize your score

Minimax is highly important in artificial intelligence for several reasons:

- It helps computers make smart decisions in two-player games
- It’s the foundation for many modern game-playing AI systems
- It teaches computers how to think ahead and plan moves

UNDERSTANDING THE MINIMAX ALGORITHM

When it's the computer's turn, it follows these steps:

- Look at all possible moves it can make
- For each move, it imagines what you might do next
- It keeps thinking ahead until it reaches an end point (like someone winning or a tie)
- It picks the move that gives it the best chance of winning

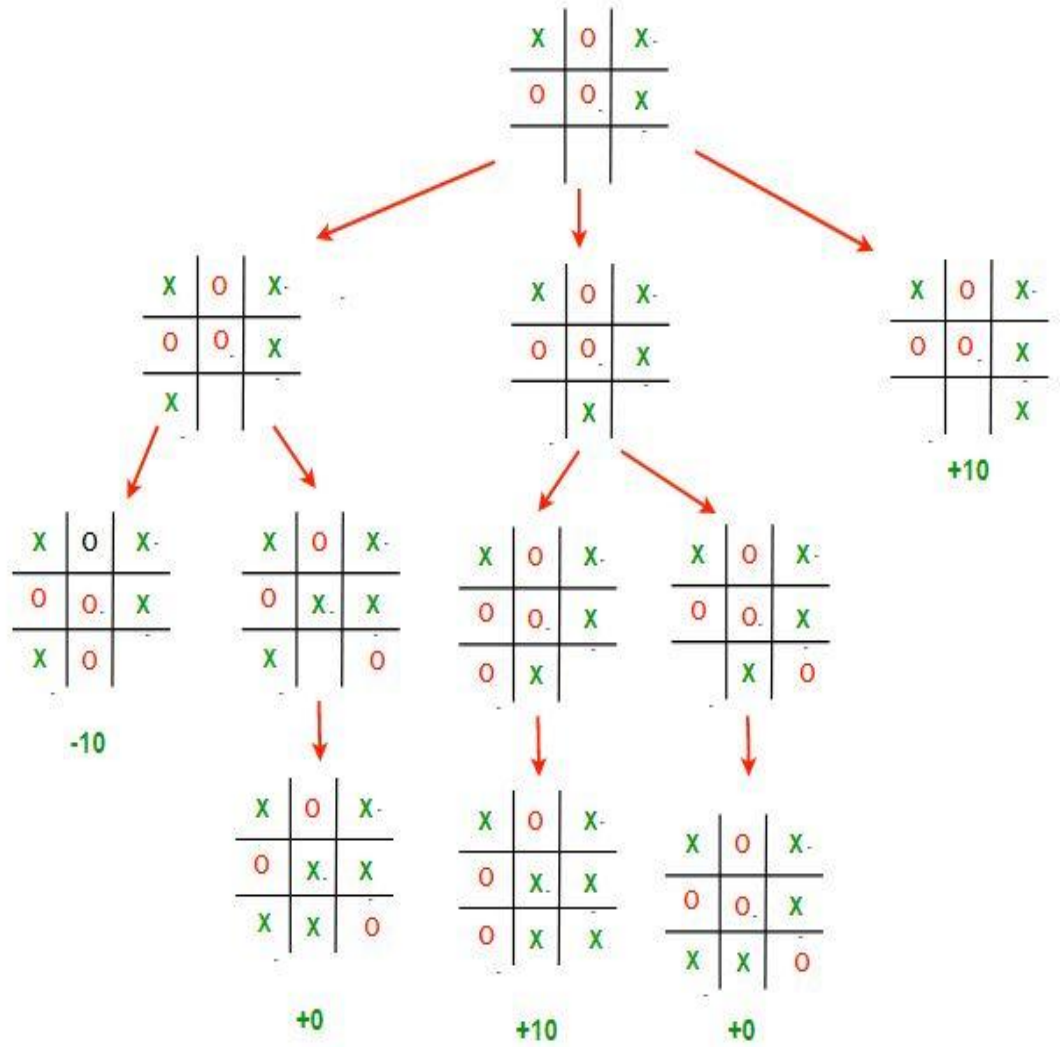
Scoring the game

The computer can use a simple scoring system:

- +10 points if the computer wins
- -10 points if you win
- 0 points if it's a tie

Think of it like a tree of possibilities:

- At the top is the current game board
- Each branch is a possible move
- At the bottom are all the possible results



APPLICATIONS OF AI

- Perception
  - Machine vision
  - Speech understanding
  - Touch (tactile or haptic) sensation
- Robotics
- Natural Language Processing
  - Natural Language Understanding
  - Speech Understanding
  - Language Generation
  - Machine Translation
- Planning
- Expert Systems
- Machine Learning
- Theorem Proving
- Symbolic Mathematics
- Game Playing